Problem 81 (Serbia TST for JBMO 2009 Problem 3)**.** There is a chess piece in each cell of an $n \times n$ chessboard where $n \geqslant 2$. In each move, we simultaneously move every chess piece to a diagonally adjacent cell. Note that each cell may contain more than one chess piece. Find the minimum number of cells which contain some chess pieces after a finite number of moves.

Solution. The answer is 4.

We label the cells in the odd rows by $1, 2, 1, 2, \ldots$, and the cells in the even rows by $3, 4, 3, 4, \ldots$. If the chess piece is put in a cell with label $1, 2, 3, 4$ respectively, it must move to a cell with label $4, 3, 2, 1$ respectively in the next step. Since there are cells of different labels at the beginning, the same holds after any number of steps. Thus, at least 4 cells contain some chess pieces.

On the other hand, we can keep on moving all chess pieces towards the top left 2×2 square. At the end, only the 4 cells in this square contain some chess pieces. So 4 is the minimum value. □

Problem 82 (European Mathematical Cup 2018 Junior Problem 1)**.** Let a, b, c be nonzero real numbers satisfying $a(a^2 + b + c) = b(b^2 + c + a) = c(c^2 + a + b) = 1$. Prove that at least two of a, b, c are equal.

Solution. Suppose on the contrary that a, b, c are pairwise distinct. Let $a + b + c = k$. Then $a(a^2 + b + c) = 1$ becomes $a(a^2 + k - a) = 1$. This means $a^3 - a^2 + ka - 1 = 0$. Similarly, as a, b, c are distinct, they are the roots to the equation $t^3 - t^2 + kt - 1 = 0$. By Vieta's theorem, we have $a + b + c = 1$. This means $k = 1$. But then the equation becomes $t^3 - t^2 + t - 1 = 0$, i.e. $(t - 1)(t^2 + 1) = 0$. It does not have 3 real roots, contradiction. □

Problem 83 (European Mathematical Cup 2016 Senior Problem 4)**.** Two circles Γ_1 and Γ_2 meet at two distinct points A and B. A line passing through A meets Γ_1 and Γ_2 again at C and D respectively. Another line passing through A meets Γ_1 and Γ_2 again at E and F respectively. Define the following intersection points.

- P is the intersection of the tangent to Γ_1 at C and the tangent to Γ_2 at D
- Q is the intersection of the tangent to Γ_1 at E and the tangent to Γ_2 at F
- S is the intersection of the tangent to Γ_1 at C and the line DF
- T is the intersection of the tangent to Γ_2 at F and the line CE
- X is the intersection of the tangent to Γ_2 at D and the line CE
- Y is the intersection of the tangent to Γ_1 at E and the line DF

Prove that the circumcircles of $\triangle BPQ$, $\triangle BST$ and $\triangle BXY$ pass through a common point different from B, or they are tangent at B.

Solution. We only work on the configuration as shown. Suppose CE meets DF at W. Let X' be the intersection point of QW and XS, and let Y' be the intersection point of PW and YT.

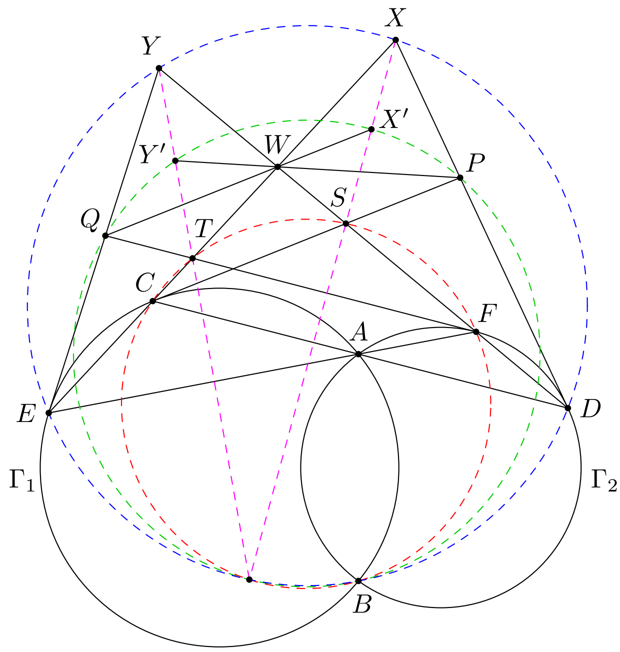
Note that B is the spiral similarity $\mathcal{T}$ that maps EF to CD. As $\angle TEF = \angle SCD$ and $\angle TFE = \angle SDC$, we have $\mathcal{T}(T) = S$. As $\angle QEF = \angle WCD$ and $\angle QFE = \angle WDC$, we have $\mathcal{T}(Q) = W$. As $\angle WEF = \angle PCD$ and $\angle WFE = \angle PDC$, we have $\mathcal{T}(W) = P$. Also, since $Y = EQ \cap FW$ and $X = CW \cap DP$, we have $\mathcal{T}(Y) = X$. In conclusion, $\mathcal{T}$ maps $EFYQTW$ to $CDXWSP$.

We now show that $\mathcal{T}(Y') = X'$. It suffices to show $\dfrac{YY'}{Y'T} = \dfrac{XX'}{X'S}$ since $\mathcal{T}$ maps YT to XS. Indeed, since $YQTW$ is similar to $XWSP$, we have

$$\begin{aligned}\frac{YY'}{Y'T} &= \frac{WY \sin \angle YWY'}{WT \sin \angle TWY'} = \frac{WY \sin \angle SWP}{WT \sin \angle XWP} = \frac{WY}{\sin \angle YQW} \cdot \frac{\sin \angle TQW}{WT} \\ &= \frac{QY}{\sin \angle YWQ} \cdot \frac{\sin \angle TWQ}{QT} = \frac{WX \sin \angle XWX'}{WS \sin \angle SWX'} = \frac{XX'}{X'S}.\end{aligned}$$

This proves our claim.

Now, as $\mathcal{T}$ maps QY' to WX', the point $Q = QY' \cap WX'$ lies on $(BY'X')$. Similarly, P lies on $(BY'X')$. Thus, (BPQ) is just $(BX'Y')$. Lastly, since B is the spiral similarity mapping $YY'T$ to $XX'S$, all the circles (BYX), $(BY'X')$, (BTS) pass through B and the intersection point of $YY'T$ and $XX'S$ by spiral similarty. $\square$



Problem 84 (USA TSTST 2015 Problem 3)**.** Let S be a nonempty set such that every element in S is a prime number. Suppose for any distinct primes $p_1, p_2, \ldots, p_k$ in S, all prime divisors of $p_1p_2\cdots p_k + 1$ belong to S. Prove that S contains all prime numbers.

Solution. Consider any prime p. Let S' be the subset of S that contains all elements a for which there are infinitely many elements $b \in S$ satisfying $a \equiv b \pmod p$. Note that there are only finitely many elements in S which do not belong to S'. Let k be the product of all elements in $S - S'$ (note that $k = 1$ if $S - S'$ is empty).

Consider the number $x_1 = k+1$. As k is a product of some elements in S, all prime divisors of $k+1$ belong to S. Also, since $q \mid k$ for any $q \in S - S'$, we have $q \nmid k+1$ for any $q \in S - S'$. Therefore, all prime divisors of $k+1$ belong to S'. Let $k+1 = p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}$ be the prime factorization. As each p_i belongs to S', we can find α_i distinct primes in S' which are congruent to p_i modulo p for $i = 1, 2, \ldots, r$. Let c be the product of all these primes, and let $x_2 = kc+1$. By construction, $x_2 \equiv kp_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r} + 1 \equiv kx_1 + 1 \pmod p$.

We repeat the process by using x_2 to replace x_1. Recursively, this generates a sequence $\{x_n\}$ such that all prime divisors of x_n belong to S, and $x_{n+1} \equiv kx_n + 1 \pmod p$. By induction, we obtain $x_n \equiv k^n + k^{n-1} + \cdots + 1 \pmod p$.

If $p \mid k$, then $p \in S - S'$, and so $p \in S$. If $k \equiv 1 \pmod p$, then $x_{p-1} \equiv 0 \pmod p$, and so $p \in S'$. If $k \not\equiv 0, 1 \pmod p$, then $x_{p-2} \equiv \dfrac{k^{p-1} - 1}{k - 1} \equiv 0 \pmod p$ by the Fermat little theorem, and so $p \in S'$. Therefore, in any case, we can deduce $p \in S$. Since this holds for every prime p, we are done. $\square$